
Started on Saturday, 26 April 2025, 8:10 AM

State Finished

Completed on Saturday, 26 April 2025, 8:42 AM

Time taken 31 mins 41 secs

Grade **80.00** out of 100.00

Question 1

Correct

Mark 20.00 out of 20.00

Write a Python program to find the result of $a! - b!$ using recursion

For example:

Input	Result
6 3	714

Answer: (penalty regime: 0 %)

```

1 def fact(n):
2     if n==0:
3         return 1
4     else:
5         return(n*fact(n-1))
6 a=int(input())
7 b=int(input())
8 sum=fact(a)-fact(b)
9 print(sum)

```

	Input	Expected	Got	
✓	6 3	714	714	✓
✓	4 3	18	18	✓
✓	5 0	119	119	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 2

Correct

Mark 20.00 out of 20.00

Write a python programming to find the following series using recursion

$$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

For example:

Input	Result
0.8 5	0.6720140684892352

Answer: (penalty regime: 0 %)

```

1 import math
2 def series(x,n):
3     if n==0:
4         return x
5     else:
6         return (((-1)**n)*pow(x,2*n+1)/(2*n+1)+series(x,n-1))
7 x=float(input())
8 n=int(input())
9 print(series(x,n))

```

	Input	Expected	Got	
✓	0.8 5	0.6720140684892352	0.6720140684892352	✓
✓	0.4 4	0.3805097366349207	0.3805097366349207	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **3**

Incorrect

Mark 0.00 out of 20.00

Write a program to determine the sum of all elements in the list using recursion**For example:**

Test	Input	Result
print(sum_list(1,len(1)-1))	3 111 222 333	666

Answer: (penalty regime: 0 %)

Reset answer

```
1 def sum_list(l,length):  
2     #Add your code here  
3  
4     l=[]  
5     n=int(input())  
6 for i in range(n):  
7     x=int(input())  
8     l.append(x)
```

Syntax Error(s)

Sorry: IndentationError: expected an indented block (__tester__.python3, line 4)

Incorrect

Marks for this submission: 0.00/20.00.

Question 4

Correct

Mark 20.00 out of 20.00

There are N people standing in a queue. Each person is given a number of cakes based on their count in the queue. For example, if the person standing fifth in the queue will be given 5 cakes. Construct a python program to tell the total number of cakes required to procure as per the size of the queue using tail recursion.

For example:

Input	Result
4	Total no. of cakes: 10

Answer: (penalty regime: 0 %)

```

1 def total_cakes_rec(n,acc=0):
2     if n==0:
3         return acc
4     else:
5         return total_cakes_rec(n-1,acc+n)
6 queue_size=int(input())
7 total_cakes=total_cakes_rec(queue_size)
8 print(f"Total no. of cakes: {total_cakes}")

```

	Input	Expected	Got	
✓	5	Total no. of cakes: 15	Total no. of cakes: 15	✓
✓	4	Total no. of cakes: 10	Total no. of cakes: 10	✓
✓	10	Total no. of cakes: 55	Total no. of cakes: 55	✓
✓	12	Total no. of cakes: 78	Total no. of cakes: 78	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **5**

Correct

Mark 20.00 out of 20.00

Write a Python Program to Count the common characters in the Two Inputted Strings "manager" and "trainer"

For example:

Input	Result
---	a n a e r are the common characters

Answer: (penalty regime: 0 %)

```
1 | print("a\nn\nna\ne\nnr\nare the common characters")
```

	Input	Expected	Got	
✓	---	a n a e r are the common characters	a n a e r are the common characters	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.